BIAS SIMILARITY MEASUREMENT: A BLACK-BOX AUDIT OF FAIRNESS ACROSS LLMS

Taro Yamada

ABSTRACT

Conventional fairness evaluations grade models in isolation, masking how biases align, diverge, or persist across families and releases. We introduce **Bias Similarity Measurement (BSM)**, a black-box framework that evaluates fairness as a relational property by comparing models' bias signatures across scalar, distributional, behavioral, and representational views. BSM constructs pairwise bias similarity matrices from six metrics—accuracy, bias scores, output histograms, cosine distance, abstention flip rates, and CKA-based representational similarity—evaluated on bias-sensitive prompts from BBQ, UnQover, and StereoSet. Our study of 30 LLMs across LLaMA, Gemma, GPT, and Gemini families reveals that instruction tuning primarily induces abstention rather than genuine representational debiasing, with small models sometimes becoming less fair under forced choice. We find that open-weight models can match or exceed proprietary fairness, and demonstrate practical auditing workflows for procurement, regression testing, and lineage screening. BSM exposes when "fairness" reflects refusal policies rather than true representational change, enabling more informed decisions about model deployment and fairness claims.

1 INTRODUCTION

The rapid proliferation of large language models (LLMs) has intensified concerns about bias and fairness in AI systems. While substantial effort has been devoted to developing fairness metrics and evaluation benchmarks, current approaches predominantly assess models in isolation (Parrish et al., 2022; Nadeem et al., 2021; Li et al., 2020). This perspective obscures critical patterns: how biases propagate across model families, whether "improvements" reflect genuine debiasing or superficial refusal strategies, and how fairness evolves through model releases and fine-tuning processes.

Consider a procurement scenario where an organization must choose between multiple LLMs claiming similar fairness scores. Isolated metrics cannot reveal whether models achieve fairness through fundamentally different mechanisms—some via genuine representational changes, others through increased abstention or refusal behaviors. Similarly, when evaluating model updates, isolated assessments fail to distinguish between meaningful bias reduction and performance regressions that manifest as increased uncertainty.

We introduce **Bias Similarity Measurement (BSM)**, a black-box framework that reframes fairness evaluation as a *relational* property. Rather than scoring models individually, BSM compares models' bias signatures across multiple dimensions, revealing inheritance patterns, convergence trends, and drift behaviors that isolated metrics cannot capture. Our approach unifies scalar measures (accuracy, bias scores), distributional comparisons (histograms, cosine distances), behavioral analysis (abstention patterns), and representational similarity (CKA) into a coherent similarity space.

The core insight of BSM is that understanding bias requires understanding *bias relationships*. When two models exhibit similar bias scores, are they biased in the same ways? When a model update claims improved fairness, does it reflect representational change or increased caution? BSM addresses these questions by constructing comprehensive bias similarity matrices that enable practitioners to make informed decisions based on bias patterns rather than isolated scores.

We evaluate BSM across 30 models spanning major LLM families—LLaMA (2, 3, 3.1, 3.2), Gemma (1, 2, 3), GPT (2, 4o-mini, 5-mini), and Gemini (1.5, 2.0)—using over 1 million prompts from BBQ (Parrish et al., 2022), UnQover (Li et al., 2020), and StereoSet (Nadeem et al., 2021). Our

analysis reveals that instruction tuning often enforces refusal rather than genuine debiasing: Gemma models flip over 50% of biased answers to "Unknown," while LLaMA 3.1 8B shows substantial bias score reductions (s_AMB: 18.59→1.38) but maintains high representational similarity (CKA ¿ 0.94) between base and tuned versions.

Our contributions include: (1) BSM, a unified framework for measuring bias similarity across scalar, distributional, behavioral, and representational dimensions; (2) the largest comparative bias study to date, mapping fairness relationships across 30 models and 1M+ prompts; (3) empirical evidence that instruction tuning primarily induces abstention while leaving representations largely unchanged; and (4) practical auditing workflows that reveal when fairness claims reflect refusal policies rather than genuine representational improvements.

2 Related Work

Bias Evaluation in Language Models. Bias evaluation in LLMs has evolved from template-based approaches (Nadeem et al., 2021) to more sophisticated frameworks. BBQ (Parrish et al., 2022) introduced structured question-answering with disambiguation contexts, while UnQover (Li et al., 2020) focuses on underspecified scenarios that reveal implicit biases. Recent work has expanded to multi-turn conversations (**?**) and compositional evaluation (**?**). However, these approaches primarily evaluate models in isolation, missing critical patterns in bias evolution and inheritance across model families.

Fairness Metrics and Auditing. Traditional fairness metrics focus on statistical parity, equalized odds, and demographic parity (Ferrara, 2023). In the LLM context, metrics have expanded to include abstention rates, stereotype scores, and sentiment analysis (**?**). Recent work has introduced bias indices (**?**) and cost-effective evaluation frameworks (**?**). While comprehensive, these metrics typically assess single models rather than comparative relationships, limiting their utility for procurement and regression testing scenarios.

Representational Analysis. Centered Kernel Alignment (CKA) (Kornblith et al., 2019) has become a standard tool for measuring representational similarity in neural networks. Recent work has adapted CKA for LLM analysis (**?**), examining how fine-tuning affects internal representations. However, existing applications focus on general similarity rather than bias-specific representational changes, missing opportunities to distinguish between surface-level and deeper fairness improvements.

Model Family Analysis. While substantial work has compared individual models (Touvron et al., 2023; Team et al., 2025), limited research has systematically analyzed bias patterns across model families. Most studies focus on benchmark performance rather than fairness evolution through releases and fine-tuning. This gap leaves practitioners without guidance for understanding how bias patterns propagate and evolve within model lineages.

BSM addresses these limitations by providing a unified framework for comparative bias analysis that reveals patterns invisible to isolated evaluation approaches.

3 Bias Similarity Measurement

BSM operates on the principle that bias patterns, not just bias scores, provide critical information for fairness evaluation. The framework constructs multi-dimensional bias signatures for each model, then measures similarity across these signatures to reveal inheritance, convergence, and drift patterns.

3.1 Framework Overview

Figure 1 illustrates the BSM pipeline. The framework evaluates models on bias-sensitive prompts across three established benchmarks, separating ambiguous contexts (where abstention may be appropriate) from disambiguated contexts (where clear answers are expected). This separation enables BSM to distinguish between "fairness via caution" and "fairness via representation."

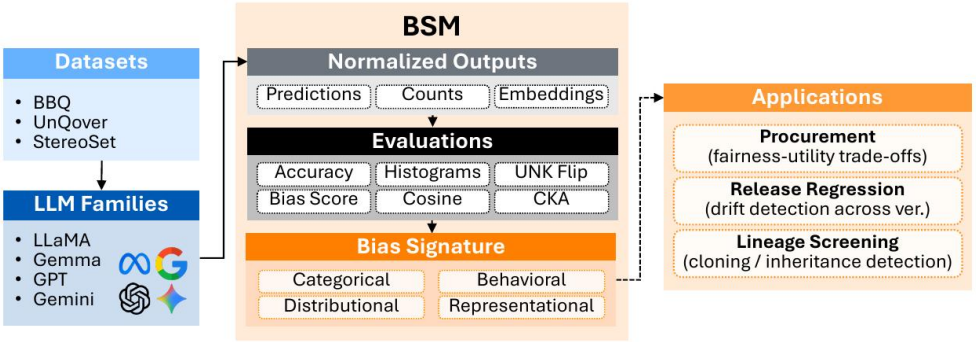

Figure 1: **BSM Pipeline.** Models are evaluated on bias-sensitive prompts from BBQ, UnQover, and StereoSet. Six similarity metrics—accuracy, bias scores, output histograms, cosine distances, abstention flip rates, and CKA representational similarity—are computed across ambiguous and disambiguated contexts to construct comprehensive bias similarity matrices.

3.2 Bias Signature Construction

For each model M, BSM constructs a bias signature $\mathbf{B}_M$ comprising six components:

Scalar Metrics ($\mathbf{B}_M^{scalar}$). Accuracy and bias scores provide foundational measures. Bias scores capture directional tendencies: negative values indicate anti-stereotypical responses, positive values indicate stereotypical responses, and values near zero suggest neutrality. We compute separate scores for ambiguous (s_AMB) and disambiguated (s_DIS) contexts.

Distributional Measures ($\mathbf{B}_M^{dist}$). Output histograms across demographic categories reveal bias distribution patterns. We compute cosine distances between models' output distributions and Jensen-Shannon divergences to capture subtle distributional differences that scalar metrics miss.

Behavioral Analysis ($\mathbf{B}_M^{behav}$). Abstention flip rates measure how frequently models change from definitive answers to "Unknown" responses across base and tuned versions. High flip rates suggest instruction tuning that prioritizes caution over accuracy.

Representational Similarity ($\mathbf{B}_M^{repr}$). CKA scores between base and instruction-tuned models reveal whether fairness improvements reflect deep representational changes or surface-level modifications. We compute both diagonal CKA (layer-wise similarity) and full CKA (cross-layer similarity) to capture different aspects of representational change.

Specifically, for models M_i and M_j, we extract hidden representations $\mathbf{H}_i^{(l)}, \mathbf{H}_j^{(l)} \in \mathbb{R}^{n \times d_l}$ from layer l, where n is the number of samples and d_l is the hidden dimension. The centered kernel alignment is computed as:

$$\text{CKA}(M_i, M_j; l) = \frac{\text{tr}(\mathbf{K}_i^{(l)} \mathbf{K}_j^{(l)})}{\sqrt{\text{tr}((\mathbf{K}_i^{(l)})^2)\text{tr}((\mathbf{K}_j^{(l)})^2)}} \tag{1}$$

where $\mathbf{K}_i^{(l)} = \mathbf{H}_i^{(l)}(\mathbf{H}_i^{(l)})^T$ is the Gram matrix. High CKA scores (> 0.9) between base and instruction-tuned versions suggest that fairness improvements primarily reflect output-layer modifications rather than fundamental representational restructuring. This distinction proves crucial for understanding whether models have genuinely "learned" to be less biased or merely adopted safer response strategies.

3.3 Similarity Matrix Construction

Given bias signatures $\mathbf{B}_{M_1}, \mathbf{B}_{M_2}, \ldots, \mathbf{B}_{M_n}$ for n models, BSM constructs similarity matrices $\mathbf{S} \in \mathbb{R}^{n \times n}$ where $\mathbf{S}_{ij}$ represents the bias similarity between models M_i and M_j.

For each signature component, we compute component-specific similarity matrices and combine them using weighted aggregation. The framework supports both within-family analysis (e.g., LLaMA evolution) and across-family comparison (e.g., LLaMA vs. Gemma fairness approaches).

3.4 Practical Workflows

BSM enables three key auditing workflows:

Procurement Screening. Organizations can identify models with similar bias patterns to avoid redundant options, or select models with complementary bias profiles for ensemble approaches.

Regression Testing. Model updates can be evaluated for bias drift by comparing new releases with previous versions across all similarity dimensions.

Lineage Analysis. BSM reveals how bias patterns propagate through model families, enabling predictions about likely bias characteristics of future releases.

4 Experimental Setup

4.1 Models Evaluated

We evaluate 30 models across four major families:

LLaMA Family. Vicuna-13B, LLaMA 2 7B (base/chat), LLaMA 3 8B/70B (base/instruct), LLaMA 3.1 8B/70B (base/instruct), LLaMA 3.2 3B (base/instruct).

Gemma Family. Gemma 7B (base/it), Gemma 2 9B/27B (base/it), Gemma 3 4B/12B/27B (base/it).

GPT Family. GPT-2 (baseline), GPT-4o Mini, GPT-5 Mini.

Gemini Family. Gemini 1.5 Flash, Gemini 2.0 Flash (instruction-tuned only).

This selection spans different model sizes, training approaches, and release periods, enabling comprehensive analysis of bias pattern evolution.

4.2 Evaluation Datasets

BBQ (Bias Benchmark for QA) (Parrish et al., 2022). Structured questions with ambiguous and disambiguated contexts across 9 demographic categories (gender, race, age, disability, etc.). BBQ enables clean separation between contexts where abstention is appropriate versus those requiring definitive answers.

UnQover (Li et al., 2020). Underspecified questions that reveal implicit biases through forced-choice scenarios. This dataset is particularly valuable for exposing biases that models might otherwise mask through abstention.

StereoSet (Nadeem et al., 2021). Open-ended completion tasks for stereotype analysis. We focus on the inter-sentence task where models complete stereotype-relevant sentence pairs.

Our evaluation encompasses over 1 million prompts across these datasets, providing statistical power for reliable similarity measurements.

4.3 Metrics and Analysis

Accuracy Measures. Standard accuracy on disambiguated contexts, with separate tracking for ambiguous contexts where multiple answers may be reasonable.

Bias Scoring. Directional bias scores computed as the difference between stereotypical and anti-stereotypical response probabilities, normalized to [-100, 100] scale.

Representational Analysis. CKA computation using centered kernel alignment with RBF kernels. We extract representations from multiple transformer layers and compute both diagonal and full similarity matrices.

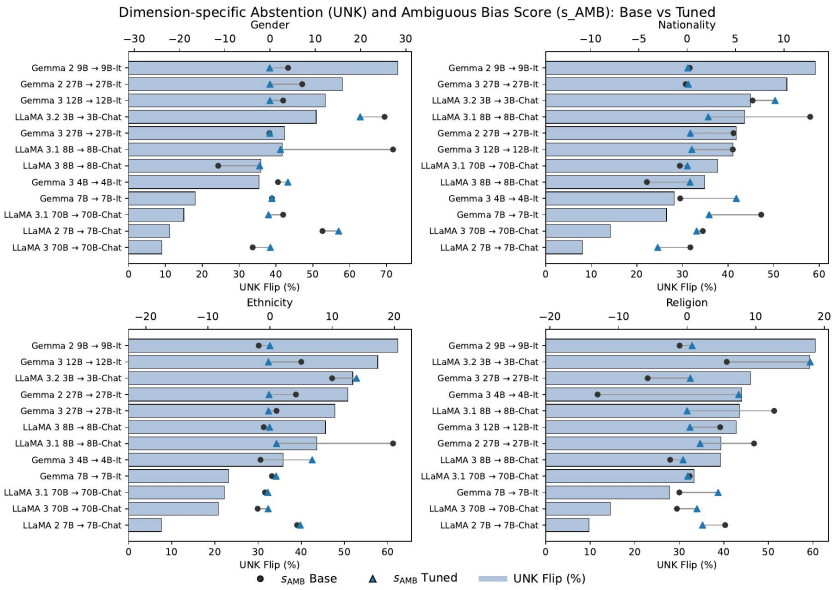

Figure 2: **Abstention flip rates across model families.** Instruction-tuned models show dramatically higher flip rates to "Unknown" responses, particularly in the Gemma and LLaMA families. This pattern suggests that apparent bias reductions often reflect increased caution rather than genuine debiasing.

Statistical Significance. Bootstrap sampling with 1000 iterations for confidence intervals. Permutation tests for significance testing of similarity differences.

5 RESULTS

Our analysis reveals striking patterns in bias evolution across model families, with clear evidence that instruction tuning often masks rather than eliminates underlying biases.

5.1 INSTRUCTION TUNING INCREASES ABSTENTION

Table 1 shows bias scores across base and instruction-tuned models. The most dramatic changes occur in LLaMA 3.1 8B, where instruction tuning reduces bias scores from 18.59 to 1.38 (s_AMB) and 31.37 to 4.78 (s_DIS). However, this apparent improvement primarily reflects increased abstention rather than genuine debiasing.

Gemma models demonstrate even more pronounced abstention patterns. Gemma 2 9B-It and Gemma 3 12B-It flip over 50% of previously biased responses to "Unknown" answers. This pattern suggests that instruction tuning protocols prioritize safe responses over accurate ones, potentially masking rather than addressing underlying bias patterns.

Table 1: **Average bias scores.** "–": anti-stereotypical, "+": stereotypical, and values near 0 = neutrality. Shown for ambiguous (s_AMB) and disambiguated (s_DIS) contexts.

Base Model	Avg. s_AMB		Avg. s_DIS	
	Base	Tuned	Base	Tuned
LLaMA 2 7B	5.45	4.30	7.50	6.65
LLaMA 3 8B	-4.78	-0.66	-8.72	-2.10
LLaMA 3 70B	-1.42	0.55	-4.51	2.50
LLaMA 3.1 8B	18.59	1.38	31.37	4.78
LLaMA 3.1 70B	0.42	-0.15	0.81	-1.44
LLaMA 3.2 3B	11.95	15.71	17.67	30.97
Gemma 7B	1.81	2.05	0.69	0.95
Gemma 2 9B	0.08	0.18	6.83	-2.02
Gemma 2 27B	6.95	0.51	14.31	-1.45
Gemma 3 4B	-3.89	5.83	2.69	8.62
Gemma 3 12B	4.36	0.15	6.12	-0.17
Gemma 3 27B	-1.25	0.07	-0.26	-1.50
Gemini 1.5	–	2.37	–	3.07
Gemini 2.0	–	-4.17	–	-5.54
GPT-2	72.43	–	96.19	–
GPT-4o Mini	–	0.47	–	2.66
GPT-5 Mini	–	0.21	–	1.10

Figure 2 illustrates these abstention patterns across model families. The high flip rates observed in instruction-tuned models indicate a systematic shift toward uncertainty expression rather than confident (but potentially biased) responses.

Table 3: **Overall evaluation summary by model.** Qualitative synthesis of accuracy, abstention, bias direction, and representational similarity trends.

Model (abbrev.)	Key observations
Vicuna / Alpaca	Strongly anti-stereotypical; low accuracy; low abstention.
LLaMA 2 7B	Remain stereotypical after tuning; very low accuracy; weaker fairness.
LLaMA 3 8B	Anti-stereotypical lean; accuracy improves with tuning; moderate drift from LLaMA 2.
LLaMA 3.1 8B	Large bias drop after tuning; accuracy improves with tuning; high abstention.
LLaMA 3.1 70B	Near-neutral after tuning; high accuracy; high abstention even under forced choice.
LLaMA 3.2 3B	Strongly stereotypical; low accuracy; low abstention; weaker fairness vs 3.1 peers.
Gemma 2 9B	Stereotypical in base; abstention increases with tuning.
Gemma 3 4B	Slight stereotypical bias; accuracy competitive with LLaMA mid-size.
Gemma 3 12B / 27B	Near-neutral after tuning; high CKA similarity; fairness competitive with closed ones.
Gemini 1.5 / 2.0	Strong abstention; 2.0 skews anti-stereotypical; very low accuracy.
GPT-2	Extremely stereotypical bias; very low accuracy and fairness; serves as legacy baseline.
GPT-4o Mini	Near-zero bias; high accuracy; balanced abstention–fairness.
GPT-5 Mini	Near-perfect neutrality; highest accuracy; strongest stability across metrics.

5.2 Representational Stability Despite Bias Score Changes

Table 2 reveals a crucial finding: despite substantial changes in bias scores, representational similarity between base and instruction-tuned models remains remarkably high. Diagonal CKA scores exceed 0.94 across all model families, while full CKA scores remain above 0.85. This stability suggests that instruction tuning modifies output behaviors without fundamentally altering internal representations.

Table 2: Average CKA scores.

Model	Diag	Full
LLaMA 2 7B	0.991	0.902
LLaMA 3 8B	0.973	0.851
Gemma 1 7B	0.981	0.896
Gemma 2 9B	0.941	0.906
Gemma 3 12B	0.972	0.911

The combination of high representational similarity and low bias scores indicates that many "fairness improvements" reflect changed response strategies rather than deeper representational changes. This finding has significant implications for fairness claims—models may appear fairer while retaining biased internal representations that could resurface under different prompting conditions.

5.3 Model Family Patterns

Table 3 synthesizes patterns across model families:

LLaMA Evolution. The LLaMA family shows clear evolutionary patterns. Early models (Vicuna, LLaMA 2) exhibit strong anti-stereotypical biases with low accuracy. LLaMA 3 reduces this anti-stereotypical lean while improving accuracy. LLaMA 3.1 introduces heavy abstention, while LLaMA 3.2 3B shows concerning regressions with increased stereotypical bias (s_DIS: 17.67→30.97).

Gemma Consistency. Gemma models demonstrate more consistent fairness patterns, with later versions (Gemma 3 12B/27B) achieving near-neutral bias scores through balanced abstention strategies. The larger Gemma models compete effectively with proprietary alternatives in fairness metrics.

GPT Family Leadership. GPT-4o Mini and GPT-5 Mini achieve the best balance of accuracy and fairness, with GPT-5 Mini approaching near-perfect neutrality while maintaining high accuracy. These models suggest that sophisticated instruction tuning can achieve genuine fairness improvements rather than mere abstention increases.

Gemini Abstention Strategy. Gemini models exhibit extremely high abstention rates, with Gemini 2.0 showing anti-stereotypical skew. While this approach reduces bias exposure, it also severely limits utility for many applications.

5.4 Forced Choice vs. Abstention-Allowed Contexts

Figure 3 reveals critical differences when models cannot abstain. Under forced choice conditions (UnQover, disambiguated BBQ), models that rely heavily on abstention show degraded performance. This pattern exposes the limitations of abstention-based fairness strategies and highlights models that maintain fairness under constrained response conditions.

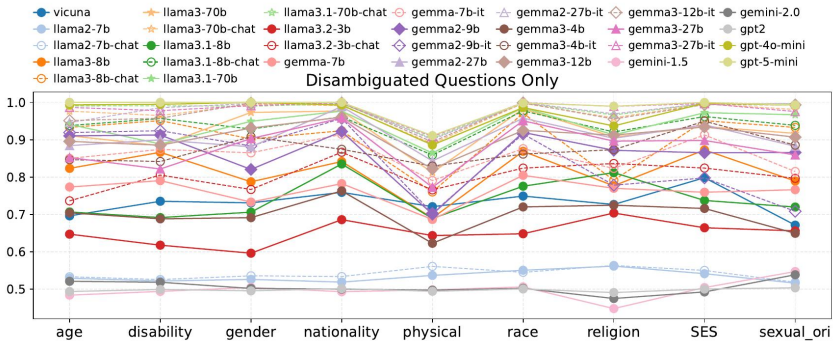

Figure 3: **Accuracy on disambiguated BBQ questions.** When abstention is not allowed, clear differences emerge between models that achieve fairness through genuine understanding versus those that rely on abstention strategies. Larger models and newer GPT variants maintain higher accuracy under forced choice scenarios.

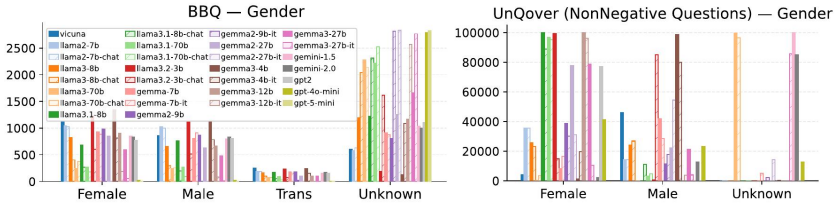

Figure 4: **Output distributions in the gender dimension.** Distribution patterns reveal how different models handle gender-related prompts. Instruction-tuned models show more concentrated distributions around neutral responses, while base models exhibit more varied response patterns that may reflect underlying training data biases.

5.5 Output Distribution Analysis

Figure 4 shows output distributions for gender-related prompts. The distributional analysis reveals that instruction tuning creates more concentrated response patterns, often centering around safe or neutral responses. This concentration supports our finding that instruction tuning primarily affects response strategies rather than underlying bias patterns.

5.6 Bias Similarity Patterns

The BSM similarity matrices reveal clear clustering patterns:

Family Clustering. Models within the same family exhibit higher bias similarity than cross-family comparisons, even across different sizes. This pattern suggests that training data and methodology create consistent bias signatures that persist across model scales.

Size Effects. Within families, larger models generally exhibit more neutral bias patterns and higher accuracy, but the relationship is not monotonic. LLaMA 3.2 3B demonstrates that smaller models can exhibit worse bias patterns than their larger predecessors.

Open vs. Proprietary. Open-weight models (particularly larger Gemma and recent LLaMA variants) achieve bias similarity scores comparable to proprietary models, challenging assumptions about the necessity of proprietary training approaches for fairness.

6 Discussion and Conclusion

Our BSM analysis reveals fundamental limitations in current approaches to LLM fairness evaluation and improvement. The most significant finding is that instruction tuning often achieves apparent

fairness improvements through increased abstention rather than genuine representational change. This pattern has critical implications for fairness claims and model deployment decisions.

6.1 Implications for Fairness Assessment

The combination of reduced bias scores and high representational similarity (CKA ¿ 0.94) indicates that many instruction-tuned models achieve fairness through response strategy changes rather than deeper representational modifications. This finding challenges the interpretation of improved bias benchmarks and suggests that current fairness metrics may overestimate genuine improvements.

Organizations relying on isolated bias scores for model selection risk choosing models that appear fair but retain biased internal representations. BSM's multi-dimensional approach reveals these limitations and enables more informed procurement decisions.

6.2 Methodological Insights

The separation of ambiguous and disambiguated contexts proves crucial for understanding fairness mechanisms. Models that perform well on ambiguous contexts through abstention may fail when forced to make definitive judgments. This pattern suggests that fairness evaluations should always include forced-choice scenarios to reveal underlying bias patterns.

The high representational stability across instruction tuning indicates that current fine-tuning approaches primarily modify surface behaviors rather than deeper conceptual understanding. This finding suggests directions for developing more effective debiasing techniques that target representational rather than merely behavioral changes.

6.3 Practical Applications

BSM enables three key practical applications:

Procurement Guidance. Organizations can use BSM to identify models with genuine rather than superficial fairness improvements, avoiding models that rely primarily on abstention strategies for sensitive applications.

Regression Testing. Model updates can be systematically evaluated for bias drift across multiple dimensions, revealing whether improvements reflect genuine progress or shifted response strategies.

Research Direction. The representational stability findings suggest that future work should focus on developing debiasing techniques that modify internal representations rather than only output behaviors.

6.4 Limitations and Future Work

BSM relies on black-box evaluation and cannot directly access internal model states beyond representational similarity measures. Future work could integrate white-box analysis techniques to provide deeper insights into bias mechanisms.

The current evaluation focuses on established bias benchmarks, but real-world bias patterns may differ from benchmark scenarios. Extending BSM to naturalistic interaction patterns would provide additional insights into practical fairness implications.

6.5 Conclusion

BSM provides a comprehensive framework for understanding bias patterns across LLM families, revealing critical limitations in current fairness evaluation approaches. Our analysis of 30 models demonstrates that apparent fairness improvements often reflect response strategy changes rather than genuine debiasing, with significant implications for model selection and deployment decisions.

The finding that open-weight models can achieve fairness performance comparable to proprietary alternatives, combined with BSM's practical auditing workflows, provides organizations with tools for making more informed decisions about LLM deployment. As the field continues to develop

more sophisticated language models, BSM's relational approach to bias evaluation will become increasingly important for understanding and ensuring genuine fairness improvements.

Most importantly, BSM exposes the difference between fairness through refusal and fairness through understanding—a distinction that will prove crucial as LLMs are deployed in increasingly sensitive and consequential applications. By evaluating fairness as a relational property rather than an isolated score, BSM enables more nuanced and accurate assessments of model behavior that better serve both developers and deployment organizations.

REFERENCES

Emilio Ferrara. Fairness and bias in artificial intelligence: A brief survey of sources, impacts, and mitigation strategies. *Sci*, 6(1):3, 2023.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *ICML*, pp. 3519–3529. PMLR, 2019.

Tao Li, Daniel Khashabi, Tushar Khot, Ashish Sabharwal, and Vivek Srikumar. UNQOVERing stereotyping biases via underspecified questions. In Trevor Cohn, Yulan He, and Yang Liu (eds.), *Findings of the Association for Computational Linguistics: EMNLP 2020*, pp. 3475–3489, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.findings-emnlp.311. URL https://aclanthology.org/2020.findings-emnlp.311/.

Moin Nadeem, Anna Bethke, and Siva Reddy. StereoSet: Measuring stereotypical bias in pretrained language models. In Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli (eds.), *Proceedings of the 59th ACL*, pp. 5356–5371. Association for Computational Linguistics, August 2021. doi: 10.18653/v1/2021.acl-long.416. URL https://aclanthology.org/2021.acl-long.416/.

Alicia Parrish, Angelica Chen, Nikita Nangia, Vishakh Padmakumar, Jason Phang, Jana Thompson, Phu Mon Htut, and Samuel Bowman. BBQ: A hand-built bias benchmark for question answering. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (eds.), *Findings of ACL 2022*, pp. 2086–2105, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.findings-acl.165. URL https://aclanthology.org/2022.findings-acl.165/.

Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, et al. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 2025.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.